

This spreadsheet contains various test cases for the functions: DGET
Name of the main junit test class which uses this spreadsheet:
`org.apache.poi.ss.formula.functions.TestDGetFunctionsFromSpreadsheet`
(The content of cell 'Read Me'!A3 is confirmed by the test)

Every sheet besides this first one contains formula evaluation test cases in a standardised format. On row 4 of each sheet, in columns B,C,D there are the column headings "Formula", "Expected Result" and "Error". The test iterates down from row 5 onward until a the text "<end>" is found in column A. Rows with "<skip>" in column A are ignored (useful for currently unsupported behaviour). Otherwise column A can be used for commenting the group of rows below.

If the evaluated result of column B does not match the expected result in column C, the junit test will report a failure. Test failures get annotated with the section and row number. Besides the first 4 columns, row 5 onwards, the junit test does not inspect any other cells in the sheet. The other cells can be used freely to set up test data.

Care should be taken to not only make the values in column C match those in column B, but also to make sure column C contains only simple literal values. This can be achieved easily by 'pasting special' from column B to C, selecting the 'as values' option.

ed Result" and "Comment"

Sheet Comment: DGET tests

	Formula	Expected Result
numeric and boolean comparisons		
	Huey	Huey
	#VALUE!	#VALUE!
	Louie	Louie
	#VALUE!	#VALUE!
	Scrooge	Scrooge
	#VALUE!	#VALUE!
	Huey	Huey
	#VALUE!	#VALUE!
	Scrooge	Scrooge
	#VALUE!	#VALUE!
	#NUM!	#NUM!
	Scrooge	Scrooge
	Scrooge	Scrooge
	Huey	Huey
string comparisons		
	Donald	Donald
	#VALUE!	#VALUE!
	Scrooge	Scrooge
	#VALUE!	#VALUE!
	Dewey	Dewey
	Dewey	Dewey
<skip>	Scrooge	Scrooge
<skip>	#VALUE!	#VALUE!
<skip>	#VALUE!	#VALUE!
<skip>	#VALUE!	#VALUE!
<skip>	Scrooge	Scrooge
<skip>	Scrooge	Scrooge
<skip>	Scrooge	Scrooge
<skip>	Scrooge	Scrooge
<skip>	#VALUE!	#VALUE!
<skip>	Scrooge	Scrooge
<skip>	Scrooge	Scrooge
<skip>	Donald	Donald
<skip>	Scrooge	Scrooge
<skip>	Scrooge	Scrooge
<skip>	#VALUE!	#VALUE!
<skip>	Dewey	Dewey
<skip>	Louie	Louie
formula comparisons		
<skip>	#VALUE!	#VALUE!
<skip>	Dewey	Dewey
<skip>	Scrooge	Scrooge
<skip>	Huey	Huey
<skip>	Donald	Donald
AND/OR		
	Louie	Louie
	#NUM!	#NUM!
	Huey	Huey

	Huey	Huey
	#NUM!	#NUM!
	Donald	Donald
corner cases matching via values		
	#VALUE!	#VALUE!
	#VALUE!	#VALUE!
	a	a
	b	b
	#VALUE!	#VALUE!
	c	c
	d	d
	d	d
	e	e
	e	e
	f	f
	f	f
	g	g
	#VALUE!	#VALUE!
	g	g
corner cases matching via formula		
	c	c
	#VALUE!	#VALUE!
	h	h
	b	b
	#VALUE!	#VALUE!
	#VALUE!	#VALUE!
	#VALUE!	#VALUE!
corner cases empty filter columns		
	a	a
	a	a
	a	a
	#VALUE!	#VALUE!
	#VALUE!	#VALUE!
numbers as condition		
	a	a
	a	a
	b	b
	b	b
strange types as headers result column		
	a	a
	a	a
	a	#VALUE!
	a	#VALUE!
	a	#VALUE!
	a	#VALUE!
	a	a
<skip>	#VALUE!	#VALUE!
	h	h
	h	h
	#VALUE!	#VALUE!
	h	h
empty cells		
	b	b

Numbers as result column

<end>

c	#VALUE!	c	#VALUE!
	#VALUE!		#VALUE!
	#VALUE!		#VALUE!
	#N/A		#N/A
	#NUM!		#NUM!
	#NAME?		#NAME?
	#DIV/0!		#DIV/0!
	#REF!		#REF!
	#NULL!		#NULL!
Text		Text	
	#VALUE!		#VALUE!
a	#VALUE!	a	#VALUE!
	#VALUE!		#VALUE!
x	#VALUE!	x	#VALUE!

Comment

smaller found
smaller not found
equals found
equals not found
larger found
larger not found
smaller equals found
smaller equals not found
larger equals found
larger equals not found
more than one result
boolean search
textual search for boolean
mixed text/number columns

equals found
equals not found
starts with found
starts with not found
starts with case insensitivity
equals case insensitivity
questionmark found
questionmark not found
questionmark does not stand for empty
questionmark does not stand for more than one
more than one questionmark in a row
more than one questionmark spread
star can be empty
star can stand for a single letter
star search failing
star can stand for multiple letters (middle)
star can stand for multiple letters (beginning)
star can stand for multiple letters (end)
multiple stars
mixing stars and questionmarks
tilde escaping tilde is broken (Excel bug)
tilde escaping questionmark
tilde escaping star

referencing database header cell
using references
using database header names (boolean comparison)
integer comparison
integer calculation and comparison

and
and (more than one result)
or (first condition hits)

database

name	age
Huey	6
Dewey	7
Louie	8
Donald	32
Scrooge	62

age	age
<7	<6

headgear	headgear
=sailor's h	=fedora

Formula	Formula
FALSE	FALSE

age	age
<30	>7

or (second condition hits)
or (more than one result)
or + and

Equality Sign, unescaped matching =
Equality Sign, escaped matching =
Equality Sign, double, escaped matching =
Empty Cell matching Empty Cell
Space matching Empty Cell
Empty Cell matching Empty Cell via Formula
Space matching Space
Empty Cell matching Space
Space matching Space via Formula
Empty Cell matching Space via Formula
#VALUE! matching #VALUE!
Empty Cell matching #VALUE!
#NUM! matching #NUM!
#VALUE! matching #NUM!
Empty Cell matching #NUM!

Index	Value
a	=
b	
c	
d	
e	
f	#VALUE!
g	#NUM!
h	abc
i	0

Empty String matching Empty String via Formula
Empty String matching Empty Cell
Empty String matching Filled Cell
Equality Sign matching Empty Cell
Equality Sign matching Filled Cell
Empty String matching 0
Equality Sign matching 0

Header: empty cell, Value: empty cell
Header: empty string, Value: empty cell
Multiple empty trailing filter columns (more than DB width)
Header: empty cell, Value: empty string
Header: empty string, Value: empty string

One
a
b

numeric condition, numeric value
textual condition, numeric value
numeric condition, textual value
textual condition, textual value

One
a
b

numeric
boolean
#VALUE!
#N/A
#NUM!
#NAME?
textual number
header not in database
header with case mismatch
result column with case mismatch
number with wrong value
unused error header values

One
a

One
def

One
a

don't fail with empty cells in searched column

don't fail with empty cells in DB headers
result cell is empty
result cell is a space
result cell is ""
#VALUE! In der Zielzelle
#N/A in der Zielzelle
#NUM! In the target cell
#NAME? in the target cell
#DIV/0! In the target cell
#REF! in the target cell
#NULL! In the target cell
Multiple matching rows, but only one non-empty
Multiple matching rows, all empty

One
a
b
c
d
e
f
g
h
i
j

Reference header by number
zero as result column
one too large as result column
textual number
textual number not found

One
a
b

size	headgear	glasses
45	cap~	FALSE
fourty-six	cap?	FALSE
47	cap*	FALSE
70	sailor's ha	FALSE
65	top hat	TRUE

age =8	age =15	age >40	age >100	age <=6	age <=5	age >=62
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headgear top	headgear fed	name dewey	name =dewey	headgear =to? hat	headgear =po? hat	headgear =to?p hat
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Formula #NAME?	Formula #NAME?	Formula #NAME?
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age <=8	glasses FALSE
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age =6 =50

age =50 =6

Equality Sign

Empty Cell

Empty String with Formula

Space

Space with Formula

Index	Value
a	=

Index	Value
c	

Index	Value
a	=

Index	Value
b	

Two	Three
c	x
d	y

One	Three
a	x

Two	
1	numeric value
2	textual value

One	Two
a	1

	2	TRUE	#VALUE!	#N/A	#NUM!	#NAME?	Three
b	c	d	e	f	g	h	

Two
a
b

One	Two
def	b

	Three
b	c

One	Three
a	c

Two	empty space =""
#VALUE!	
#N/A	
#NUM!	
#NAME?	
#DIV/0!	
#REF!	
#NULL!	

One a

One b

One	Two
a	
a	Text
a	
b	
b	

One a

Two		1
c	x	
d	y	

One a

age	age	glasses	glasses	size
>=100	<10	TRUE	=TRUE	<47

headgear	headgear	headgear	headgear	headgear	headgear	headgear
=t?hat	=t???hat	=t?p?h?t	=to*p hat	=t*p hat	=toph*	=t*t

age
<7
<10

age	size
=62	=60
=32	=70

Index	Value
a	==

Index	Value
b	

Index	Value
h	

Index	Value
b	=

One	Three
a	x

One
a

One	Two
a	1

One	Two
b	2

One
b

One		2
a	b	

One	TRUE
a	c

One	2
a	b

One	3
a	b

one
a

One
a

One	2
a	c

One
c

One
d

One
e

One
f

One
b

headgear	headgear	headgear	headgear	headgear	headgear	headgear
p hat	=sailo	=*p h*t	=t?p h*	=cap ^{^^}	=cap [~] ?	=cap [~] *

Index	Value
b	

Index	Value
h	=

Index	Value
c	

Index	Value
i	

Index	Value
d	

Index	Value
i	=

One	Three
a	x

One
a

Two
2

One	#VALUE!
a	d

One	#N/A
a	e

One	#NUM!
a	f

One
g

One
h

One
i

One
j

Index	Value
d	

Index	Value
e	

Index	
e	

Three
x

One	#NAME?
a	g

One	Three
a	h

Value

Index	Value
f	#VALUE!

Index	Value
f	

Index	Value
g	#NUM!

Index	Value
g	#VALUE!

Index	Value
g	